Point Defects in Crystals from Freysoldt *et al.*

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1 The Formation Energy as a Grand Potential

Reservoirs and the grand potential. A crystal supercell exchanges atoms of each species i with a reservoir at the chemical potential μ_i, and exchanges electrons with a reservoir at the electron chemical potential, the Fermi level E_F. The reservoirs represent the surroundings of the supercell. They fix the temperature T and the chemical potentials, and the supercell has a fixed volume V. The Legendre transform of the internal energy U with the natural variables $(T, V, \{\mu_i\}, E_F)$ is the grand potential Φ. Here, N_i denotes the number of atoms of species i and N_e the number of electrons in the pristine supercell. The grand potential of the pristine supercell is

$$\Phi_{\text{bulk}} = U_{\text{bulk}} - TS_{\text{bulk}} - \sum_i \mu_i N_i - E_F N_e = F_{\text{bulk}} - \sum_i \mu_i N_i - E_F N_e.$$

Neutral defects. Here, n_i denotes the number of atoms of species i that enter the supercell to form the defect X. A positive n_i adds atoms, for example an interstitial, and a negative n_i removes atoms, for example a vacancy. The grand potential of the supercell with the defect is

$$\Phi[X] = F[X] - \sum_i \mu_i \left(N_i + n_i\right) - E_F N_e.$$

The formation energy of the defect is the change of the grand potential at fixed $(T, V, \{\mu_i\}, E_F)$,

$$\boxed{E^f[X] = \Phi[X] - \Phi_{\text{bulk}} = F[X] - F_{\text{bulk}} - \sum_i \mu_i n_i.} \tag{1.1}$$

The natural variables are the same before and after the defect forms, but $\Phi[X]$ and Φ_{bulk} describe two different systems, so the two values differ.

Charged defects. A defect in the charge state q, written X^q, has released q electrons to the electron reservoir, so the supercell holds $N_e - q$ electrons,

$$\Phi[X^q] = F[X^q] - \sum_i \mu_i \left(N_i + n_i\right) - E_F \left(N_e - q\right).$$

The terms in N_i and N_e cancel against Φ_{bulk}, and the formation energy is

$$\boxed{E^f[X^q] = F[X^q] - F_{\text{bulk}} - \sum_i \mu_i n_i + qE_F.} \tag{1.2}$$

Formation energy from DFT at 0 K. At $T = 0$, the free energy equals the internal energy, $F = U$. A DFT calculation gives the total energy E_{tot} of the supercell with the nuclei at rest. The nuclei still vibrate at $T = 0$, so the internal energy also contains the zero-point energy of the phonon modes. Here, $E_{\text{ZPE}} = \frac{1}{2}\sum_k \hbar\omega_k$ denotes the zero-point energy of a supercell with the phonon frequencies ω_k. With $U = E_{\text{tot}} + E_{\text{ZPE}}$, the formation energy reads

$$\boxed{E^f[X^q] = \big(E_{\text{tot}}[X^q] + E_{\text{ZPE}}[X^q]\big) - \big(E_{\text{tot}}[\text{bulk}] + E_{\text{ZPE}}[\text{bulk}]\big) - \sum_i \mu_i n_i + qE_F.} \tag{1.3}$$

The total energies come from the DFT calculations, and the zero-point energies from the phonon frequencies of the two supercells. The phonon frequencies scale as $M^{-1/2}$ with the nuclear mass M, and most modes are the same in both supercells, so the difference of the zero-point energies is negligible for heavy nuclei and matters mainly for light atoms such as hydrogen. The chemical potentials μ_i and the Fermi level E_F describe the reservoirs and enter as parameters.

2 Charge-State Transition Levels and the U Parameter

The formation energy as a function of the Fermi level. Here, $E_0^f[X^q]$ denotes the formation energy at $E_F = 0$, with E_F measured from the valence-band maximum (VBM). The formation energy of Section 1 depends on the Fermi level through the term qE_F only,

$$E^f[X^q] = E_0^f[X^q] + qE_F.$$

In a plot against E_F, each charge state is a straight line with the slope q. At a given E_F, the defect takes the charge state with the lowest formation energy, so the stable charge state follows the lower envelope of the lines (red in Fig. 1). The Fermi level ranges from the VBM to the conduction-band minimum (CBM).

Charge-state transition levels. The charge-state transition level $\varepsilon(q/q')$ is the Fermi level at which the charge states q and q' have equal formation energies. Equating the two lines gives

$$\boxed{\varepsilon(q/q') = \frac{E_0^f[X^{q'}] - E_0^f[X^q]}{q - q'}.} \tag{2.1}$$

Below $\varepsilon(q/q')$, the more positive charge state is stable, and above $\varepsilon(q/q')$, the more negative one is stable. The transition levels are the kinks of the lower envelope.

The stability window of a charge state. Three consecutive charge states $q+1$, q and $q-1$ have two transition levels, $\varepsilon(q+1/q) = E_0^f[X^q] - E_0^f[X^{q+1}]$ and $\varepsilon(q/q-1) = E_0^f[X^{q-1}] - E_0^f[X^q]$. The charge state q is stable only for Fermi levels between the two. The width of this window defines the U parameter,

$$\boxed{U \equiv \varepsilon(q/q-1) - \varepsilon(q+1/q) = E^f[X^{q+1}] + E^f[X^{q-1}] - 2E^f[X^q].} \tag{2.2}$$

The terms in E_F cancel, because $(q+1) + (q-1) - 2q = 0$, so U does not depend on the Fermi level.

U as a reaction energy. The right-hand side of the definition is the energy change of the disproportionation reaction

$$2X^q \to X^{q+1} + X^{q-1},$$

in which one defect passes an electron to another. The electron does not reach the reservoir, which explains why E_F drops out.

The sign of U. A second electron on the same defect feels the Coulomb repulsion of the first, which favors $U > 0$. The charge state q is then stable in a window of width U (Fig. 1a). The lattice around the defect can relax differently in each charge state, and a large relaxation can outweigh the repulsion, so that $U < 0$. The disproportionation then releases energy, the line of the charge state q lies above the crossing of the lines $q+1$ and $q-1$ for every E_F, and the defect changes directly from $q+1$ to $q-1$ at the level $\varepsilon(q+1/q-1)$ (Fig. 1b). The vacancy in silicon is an example, with an unstable V^+.

More charge states. With more than three charge states, the lower envelope selects the stable ones in order of decreasing charge from the VBM to the CBM. Each charge state that appears has a window of its own width, and a charge state with a negative U drops out, so that the envelope jumps by two units of slope at a single kink (Fig. 1c).

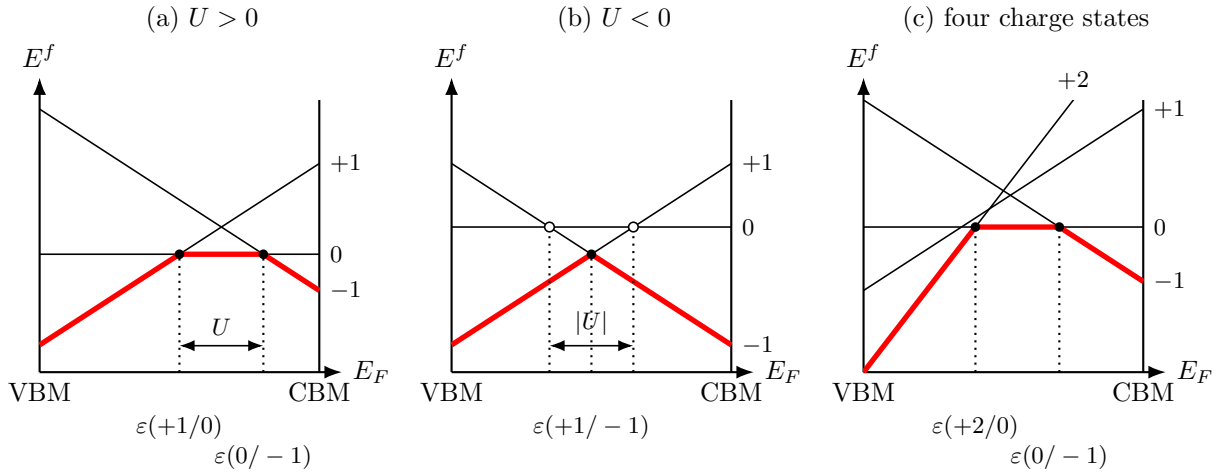


Figure 1: Formation energy against the Fermi level for three model defects, with the lower envelope in red. (a) Charge states +1, 0 and −1 with $U > 0$ and a window for the neutral state. (b) The same charge states with $U < 0$. Open circles: crossings with the neutral line above the envelope. (c) Charge states +2, +1, 0 and −1, with the state +1 skipped and a direct level $\varepsilon(+2/0)$.

3 Configurational Entropy and the Equilibrium Concentration

Counting the defect arrangements. Here, N_X denotes the number of defects in the crystal, to keep it distinct from the composition n_i of one defect in Section 1. A crystal has N sites, and N_X defects occupy them. Each defect has g_X internal orientations. The problem is to count the ways to put N_X defects on N sites, with one defect per site and g_X orientations for each defect. The i-th defect has $N-(i-1)$ free sites and g_X orientations, so the number of ways to place N_X distinguishable defects in the supercell is

$$W = \prod_{i=1}^{N_X} g_X\,(N-i+1) = g_X^{N_X}\,\frac{N!}{(N-N_X)!}.$$

For large N, the Stirling approximation

$$x! \approx \sqrt{2\pi x}\left(\frac{x}{e}\right)^x$$

gives

$$\frac{N!}{(N-N_X)!} \approx \sqrt{\frac{N}{N-N_X}}\,\frac{N^N}{(N-N_X)^{N-N_X}}\,e^{-N_X}.$$

The case of interest is a very dilute defect concentration. The middle factor is

$$\frac{N^N}{(N-N_X)^{N-N_X}} = N^{N_X}\left(\frac{N}{N-N_X}\right)^{N-N_X} = N^{N_X}\left(1-\frac{N_X}{N}\right)^{-(N-N_X)} \approx N^{N_X}e^{N_X},$$

which cancels the factor e^{-N_X}, and the square root tends to one, so

$$\lim_{N_X/N\to 0} g_X^{N_X}\sqrt{\frac{N}{N-N_X}}\,\frac{N^N}{(N-N_X)^{N-N_X}}\,e^{-N_X} = (g_X N)^{N_X}.$$

Identical defects can be permuted without changing the configuration. Each distinct configuration has $N_X!$ permutations, so the number of ways to put N_X indistinguishable defects into the crystal is

$$\boxed{W = g_X^{N_X}\,\frac{N!}{(N-N_X)!\,N_X!} \approx \frac{(g_X N)^{N_X}}{N_X!}.} \tag{3.1}$$

Configurational entropy per site. The configurational entropy of the defect system at fixed internal energy, crystal volume and particle number is

$$S_{\text{conf}} = k_B \ln W.$$

The Stirling formula

$$\ln N_X! \approx N_X \ln N_X - N_X$$

gives

$$S_{\text{conf}} = k_B \big[N_X \ln(g_X N) - N_X \ln N_X + N_X \big].$$

The defect concentration is

$$c = \frac{N_X}{N},$$

and the entropy as a function of c is

$$\begin{aligned} S_{\text{conf}} &= k_B \big[cN \ln(g_X N) - cN \ln(cN) + cN \big] \\ &= N k_B \big[c \ln g_X - c \ln c + c \big]. \end{aligned}$$

The entropy is proportional to N, so S_{conf} is extensive, and dividing by N expresses the entropy per site. Here, s_{conf} denotes the configurational entropy per site,

$$\boxed{s_{\text{conf}}(c) = k_B \big[c \ln g_X - c \ln c + c \big].} \tag{3.2}$$

Without the dilute limit, the Stirling formula applied to every factorial of the exact W gives

$$\begin{aligned} \ln W &\approx N_X \ln g_X + N \ln N - (N - N_X) \ln(N - N_X) - N_X \ln N_X \\ &= N \big[c \ln g_X - c \ln c - (1 - c) \ln(1 - c) \big], \end{aligned}$$

so the exact entropy per site is

$$s_{\text{conf}}(c) = k_B \big[c \ln g_X - c \ln c - (1 - c) \ln(1 - c) \big].$$

For $c \ll 1$, the last term is $-(1 - c) \ln(1 - c) \approx c$, and the dilute result follows.

Restoring the temperature at fixed pressure. Section 1 neglected the temperature dependence of the defect formation energy by setting the temperature to 0 K in the DFT calculation. This section restores the temperature, with the assumption that the internal energy difference between the defective and the pristine crystal does not depend on the temperature. The reservoirs fix the chemical potentials in both the grand-canonical description of Section 1 and in experiments. The difference is that experiments fix the pressure P instead of the volume, so the crystal volume changes when a defect forms. The natural thermodynamic potential is therefore the Gibbs free energy. Here, V_{bulk} and $V[X^q]$ denote the volumes of the pristine and the defective crystal at the pressure P, and ΔV the difference,

$$\Delta V = V[X^q] - V_{\text{bulk}}.$$

The Gibbs free energies are

$$G_{\text{bulk}} = \Phi_{\text{bulk}} + \sum_i \mu_i N_i + E_F N_e + P V_{\text{bulk}},$$
$$G[X^q] = \Phi[X^q] + \sum_i \mu_i \left(N_i + n_i\right) + E_F \left(N_e - q\right) + P V[X^q].$$

The formation energy at fixed pressure is

$$\begin{aligned} E^f[X^q] &= G[X^q] - G_{\text{bulk}} - \sum_i \mu_i n_i + q E_F \\ &= F[X^q] - F_{\text{bulk}} - \sum_i \mu_i n_i + q E_F + P\,\Delta V, \end{aligned}$$

which is the formation energy of Section 1 plus the term $P\,\Delta V$. At ambient pressure, this term is

$$P\,\Delta V \approx 10^5 \text{ Pa} \times 10^{-29} \text{ m}^3 \approx 10^{-5} \text{ eV},$$

which is negligible against formation energies of the order of 1 eV, so the formation energy of Section 1 also holds at fixed pressure. At finite temperature, the change of the Gibbs free energy includes the change in entropy of the defect and the bulk system. The entropy has three contributions, the electronic, the vibrational and the configurational entropy. Only the configurational entropy is considered here. Here, $G(N_X)$ denotes the Gibbs free energy of the crystal with N_X defects, and $\Delta G(N_X)$ the change of the Gibbs free energy without the reservoir terms,

$$\Delta G(N_X) = G(N_X) - G_{\text{bulk}} - N_X \Big(\sum_i \mu_i n_i - q E_F \Big).$$

Each defect adds $E^f[X^q]$, and the arrangement of all N_X defects adds the configurational entropy, so

$$\boxed{\Delta G(N_X) = N_X\, E^f[X^q] - T\, S_{\text{conf}}.} \tag{3.3}$$

Equilibrium defect concentration. Here, Δg denotes the change of the Gibbs free energy per site,

$$\Delta g(c) = \frac{\Delta G(N_X)}{N} = c\, E^f[X^q] - T\, s_{\text{conf}}(c).$$

Defects form and disappear freely, so the concentration c adjusts until Δg is a minimum. With the dilute s_{conf}, the condition for the minimum is

$$\begin{aligned} \frac{\partial \Delta g}{\partial c} &= E^f[X^q] - T\, \frac{\partial s_{\text{conf}}}{\partial c} \\ &= E^f[X^q] - k_B T \big[\ln g_X - \ln c - 1 + 1 \big] \\ &= E^f[X^q] - k_B T \big[\ln g_X - \ln c \big] = 0, \end{aligned}$$

which gives the equilibrium concentration

$$\boxed{c_{\text{eq}} = g_X \, e^{-E^f[X^q]/k_B T}.} \tag{3.4}$$